QUBO Formulation: RNA Secondary-Structure Stem Selection

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Problem Statement. Given a set of N candidate stems indexed by $i \in \{0, 1, \ldots, N-1\}$, each associated with a positive stability score $w_i > 0$, and a set of pairwise conflict pairs $C \subseteq \{0, 1, \ldots, N-1\} \times \{0, 1, \ldots, N-1\}$, the goal is to select a subset of stems that maximizes the total stability score while ensuring that no two conflicting stems are selected simultaneously.

Decision Variables. For each candidate stem $i \in \{0, 1, \ldots, N-1\}$, introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i takes the value 1 if stem i is selected, and 0 otherwise.

QUBO Derivation. **Step 1.** The natural objective is to maximize the total stability score:

$$\max_x \sum_{i=0}^{N-1} w_i x_i. \tag{1}$$

Since QUBO solvers minimize energy, we negate the objective to obtain a minimization problem:

$$\min_x \left(-\sum_{i=0}^{N-1} w_i x_i \right). \tag{2}$$

Step 2. The compatibility constraints require that for every conflicting pair $(i, j) \in C$, the stems cannot both be selected:

$$x_i + x_j \leq 1, \quad \forall (i, j) \in C. \tag{3}$$

Step 3. Each inequality constraint is converted to a quadratic penalty term. For a pair $(i, j) \in C$, the squared penalty is:

$$\lambda (x_i + x_j - 1)^2. \tag{4}$$

Expanding the square yields:

$$\lambda(x_i^2 + x_j^2 + 2x_i x_j - 2x_i - 2x_j + 1). \tag{5}$$

Applying the idempotent property $x_k^2 = x_k$ for binary variables gives:

$$\lambda(x_i + x_j + 2x_i x_j - 2x_i - 2x_j + 1) = \lambda(1 - x_i - x_j + 2x_i x_j). \tag{6}$$

Constant terms do not affect the minimizer, and linear terms can be absorbed into the objective. The essential quadratic contribution that enforces exclusivity on $\{0,1\}^N$ is $\lambda x_i x_j$. Summing over all conflicts, the total penalty contribution to the Hamiltonian is:

$$\lambda \sum_{(i,j)\in C} x_i x_j. \tag{7}$$

Step 4. Combining the negated objective and the penalty terms yields the unconstrained QUBO Hamiltonian $H(x)$:

$$H(x) = -\sum_{i=0}^{N-1} w_i x_i + \lambda \sum_{(i,j)\in C} x_i x_j. \tag{8}$$

Step 5. Reading off the coefficients from $H(x) = \sum_{i,j} Q_{ij} x_i x_j + c$, the Q matrix entries and offset are given in closed form as:

$$Q_{i,i} = -w_i, \qquad i \in \{0, \ldots, N-1\}, \tag{9}$$
$$Q_{i,j} = \lambda, \qquad (i,j) \in C, \tag{10}$$
$$Q_{j,i} = \lambda, \qquad (i,j) \in C, \tag{11}$$
$$Q_{i,j} = 0, \qquad \text{otherwise}, \tag{12}$$
$$c = 0. \tag{13}$$

Penalty Weight Justification. The penalty weight λ must satisfy $\lambda > \max_{i\in\{0,\ldots,N-1\}} w_i$. This condition guarantees that the energy cost of violating any single conflict constraint (which is at least λ) strictly exceeds the maximum possible gain in the objective function from selecting any individual stem (which is at most $\max_i w_i$). Consequently, the optimizer will never prefer an infeasible assignment over a feasible one.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^N$ recovers the optimal solution because (i) for any feasible assignment satisfying all constraints in C, every penalty term vanishes, reducing $H(x)$ exactly to the negated original objective; thus, minimizing H is equivalent to maximizing the total stability score. (ii) For any infeasible assignment, at least one conflict constraint is violated, incurring a penalty of at least λ. By the choice of λ, this penalty strictly dominates the maximum achievable objective value. Therefore, the global minimum of $H(x)$ must correspond to a feasible assignment that maximizes the original objective.

Illustrative Example. Consider a small instance with $N = 3$ candidate stems. Let the stability scores be $w_0 = 3$, $w_1 = 5$, and $w_2 = 4$. The conflict set is $C = \{(0,1),(1,2)\}$. Following the general formulation, we choose $\lambda > \max\{3,5,4\}$, so we set $\lambda = 6$.

Instantiating the general Hamiltonian yields:

$$H(x) = -3x_0 - 5x_1 - 4x_2 + 6x_0x_1 + 6x_1x_2. \tag{14}$$

The corresponding Q matrix entries are:

$$\begin{aligned} &Q_{0,0} = -3, && Q_{1,1} = -5, && Q_{2,2} = -4, && (15)\\ &Q_{0,1} = Q_{1,0} = 6, && Q_{1,2} = Q_{2,1} = 6, && && (16)\\ &Q_{i,j} = 0 \quad \text{for all other } (i,j), c && && = 0. && (17) \end{aligned}$$

Evaluating $H(x)$ over all $2^3 = 8$ binary assignments:

x	Feasible?	$H(x)$
$(0,0,0)$	Yes	0
$(1,0,0)$	Yes	-3
$(0,0,1)$	Yes	-4
$(1,0,1)$	Yes	-7
$(0,1,0)$	No	-5
$(1,1,0)$	No	-2
$(0,1,1)$	No	-3
$(1,1,1)$	No	0

(18)

The global minimum is $H(x^*) = -7$ at $x^* = (1,0,1)$. This corresponds to selecting stems 0 and 2, which yields a maximum total stability score of $3 + 4 = 7$. The penalty terms correctly suppress all infeasible configurations, confirming the formulation.